Mathematics - Undergraduate Level Assessment 4 Total Marks: 40 Answer all questions.

Multiple Choice (18 marks)

1. If c > 0 and f(x) = e^x - cx for all real numbers x, then the minimum value of f is
   - (a) f(c)
   - (b) f(e^c)
   - (c) f(1/c)
   - (d) f(log c)
2. If A is a subset of the real line R and A contains each rational number, which of the following mus
   - (a) If A is open, then A = R.
   - (b) If A is closed, then A = R.
   - (c) If A is uncountable, then A = R.
   - (d) If A is uncountable, then A is open.
3. Statement 1 | Every permutation is a one-to-one function. Statement 2 | Every function is a perm
   - (a) True, True
   - (b) False, False
   - (c) True, False
   - (d) False, True
4. Find the characteristic of the ring Z x Z.
   - (a) 0
   - (b) 3
   - (c) 12
   - (d) 30
5. Find all c in Z_3 such that Z_3[x]/(x^3 + x^2 + c) is a field.
   - (a) 0
   - (b) 2
   - (c) 1
   - (d) 3
6. The shortest distance from the curve xy = 8 to the origin is
   - (a) 4

(b) 8
(c) 16
(d) 2sqrt(2)

7. If f is a linear transformation from the plane to the real numbers and if f(1, 1) = 1 and f(-1, 0) = 2,

   (a) 9
   (b) 8
   (c) 0
   (d) -5

8. Find the characteristic of the ring Z_3 x Z_3.

   (a) 0
   (b) 3
   (c) 12
   (d) 30

9. Find the generator for the finite field Z_7.

   (a) 1
   (b) 2
   (c) 3
   (d) 4

True / False (2 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'The polynomial x^3 + 2x^2 + 2x + 1 can be factored into lir
   Find this factorization.' is '(x + 1)(x - 4)(x - 2)'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Simplify
   $$\frac{bkw\sin x - \cos x}{cmb\sin x + \cos x}n$$
   The answer will be a trigonometric function of some simple function of $x,$ like "$\cos 2x$" or "$\sin(x^3)$".

2. What is the smallest possible number of whole 2-by-3 non-overlapping rectangles needed to cover a square region exactly, without extra over-hangs and without gaps?

End of Paper